

UNIT-2

Two-Dimensional Random Variable

Joint probability mass function of discrete variables.

Let (X, Y) be a two dimensional random discrete variables with values $(x_i, y_j, i=1, 2, \dots, j=1, 2, \dots)$

i) $P(x_i, y_j) \geq 0$ for all $i=1, 2, \dots$ & $j=1, 2, \dots$

ii) $\sum_i \sum_j P(x_i, y_j) = 1$

Joint probability density function of continuous variables.

Let (X, Y) be a continuous two dimensional random variables. A function 'f' which assigns each (x, y) a real number $f(x, y)$ for all real x, y is called a joint probability density function of (X, Y) if.

i) $f(x, y) \geq 0$, for all x & y in R

ii) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

Joint Cumulative distribution function:

i) For discrete case

$$F_{xy}(x, y) = \sum_j \sum_i P_{ij}$$

ii) For continuous case

$$F_{xy}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx$$

Marginal Probability Density Function:

i) For discrete,

$$P_x(x_i) = \sum_{j=1}^{\infty} P(x_i, y_j), \quad i=1, 2, 3, \dots$$

$$P_y(y_j) = \sum_{i=1}^{\infty} P(x_i, y_j), \quad j=1, 2, \dots$$

ii) For continuous.

$$f_x(x) = \int_{-\infty}^{\infty} f_{xy}(x, y) dy$$

$$f_y(y) = \int_{-\infty}^{\infty} f_{xy}(x, y) dx$$

Conditional probability Distribution Function:

$$f_y(Y/x) = \frac{f_{xy}(x, y)}{f_x(x)}, \quad f_x(x) > 0$$

i) Discrete

$$P_x(x_i/y_j) = \frac{P(x=x_i, y=y_j)}{P(y=y_j)} = \frac{P(x_i, y_j)}{P_y(y_j)}$$

$$P_y(y_j/x_i) = \frac{P(x=x_i, y=y_j)}{P(x=x_i)} = \frac{P(x_i, y_j)}{P_x(x_i)}$$

ii). For continuous

$$f_y(y/x) = \frac{f_{xy}(x,y)}{f_x(x)}$$

$$f_x(x/y) = \frac{f_{xy}(x,y)}{f_y(y)}$$

①. For the bivariate probability distribution of (x,y) given below, find $P(x \leq 1)$, $P(y \leq 3)$, $P(x \leq 1, y \leq 3)$, $P(x \leq 1 / y \leq 3)$, $P(y \leq 3 / x \leq 1)$ and $P(x+y \leq 4)$

$x \backslash y$	1	2	3	4	5	6
0	0	0	$1/32$	$2/32$	$2/32$	$3/32$
1	$1/16$	$1/16$	$1/8$	$1/8$	$1/8$	$1/8$
2	$1/32$	$1/32$	$1/64$	$1/64$	0	$2/64$

Sol.

$$P(x \leq 1) = P(x=0) + P(x=1)$$

$$= \left(\frac{1}{32} + \frac{2}{32} + \frac{2}{32} + \frac{3}{32} \right) + \left(\frac{2}{16} + \frac{4}{8} \right)$$

$$= \frac{18}{32} + \frac{12+8}{16}$$

$$= \frac{1}{4} + \frac{10}{16} = \frac{14}{16}$$

$$= \frac{7}{8}$$

$$\begin{aligned}
 P(Y \leq 3) &= P(Y=1) + P(Y=2) + P(Y=3) \\
 &= \left(\frac{1}{16} + \frac{1}{32}\right) + \left(\frac{1}{16} + \frac{1}{32}\right) + \left(\frac{1}{32} + \frac{1}{8} + \frac{1}{64}\right) \\
 &= \frac{2}{16} + \frac{3}{32} + \frac{1}{8} + \frac{1}{64} \\
 &= \frac{2}{8} + \frac{3}{32} + \frac{1}{64} \\
 &= \frac{16+6+1}{64} \\
 &= \frac{23}{64}
 \end{aligned}$$

$$\begin{aligned}
 P(X \leq 1, Y \leq 3) &= \sum_{j=1}^3 P(X=0, Y=y_j) + \sum_{j=1}^3 P(X=1, Y=y_j) \\
 &= P(X=0, Y=1) + P(X=0, Y=2) + P(X=0, Y=3) + \\
 &\quad P(X=1, Y=1) + P(X=1, Y=2) + P(X=1, Y=3) \\
 &= 0 + 0 + \frac{1}{32} + \frac{1}{16} + \frac{1}{16} + \frac{1}{8} \\
 &= \frac{2}{16} + \frac{1}{32} + \frac{1}{8} \\
 &= \frac{4+1+4}{32} \\
 &= \frac{9}{32}
 \end{aligned}$$

$$\begin{aligned}
 P(X \leq 1 | Y \leq 3) &= \frac{P(X \leq 1, Y \leq 3)}{P(Y \leq 3)} \\
 &= \frac{9/32}{23/64} \\
 &= \frac{18}{23}
 \end{aligned}$$

$$P(Y \leq 3 | X \leq 1) = \frac{P(Y \leq 3, X \leq 1)}{P(X \leq 1)}$$

$$= \frac{9/32}{7/8}$$

$$= \frac{9}{28}$$

$$P(X+Y \leq 4) = \sum_{j=1}^4 P(X=0, Y=y_j) + \sum_{j=1}^3 P(X=1, Y=y_j) + \sum_{j=1}^2 P(X=2, Y=y_j)$$

$$= 0 + 0 + \frac{1}{32} + \frac{2}{32} + \frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{32} + \frac{1}{32}$$

$$= \frac{5}{32} + \frac{2}{16} + \frac{1}{8}$$

$$= \frac{5}{32} + \frac{2}{8}$$

$$= \frac{5+8}{32}$$

$$= \frac{13}{32}$$

- ⑤ The joint probability mass fn of (X, Y) is given by $P(X, Y) = k(2x+3y)$, $x = 0, 1, 2$, $y = 1, 2, 3$. Find all the marginal and conditional probability distribution. also find the probability distribution of $X+Y$.

Sol:

Given, $P(X, Y) = k(2x+3y)$

$x = 0, 1, 2$

$y = 1, 2, 3$

$x \backslash y$	1	2	3
0	$3k$	$6k$	$9k$
1	$5k$	$8k$	$11k$
2	$7k$	$10k$	$13k$

To find k :

$$\sum_{j=1}^3 \sum_{i=0}^2 P_{ij} = 1$$

$$\Rightarrow k = 1$$

$$k = 1/42$$

$x \backslash y$	1	2	3	P_{X_i}
0	$3/42$	$6/42$	$9/42$	$\frac{18}{42}$
1	$5/42$	$8/42$	$11/42$	$\frac{24}{42}$
2	$7/42$	$10/42$	$13/42$	$\frac{30}{42}$
P_{Y_j}	$\frac{15}{42}$	$\frac{24}{42}$	$\frac{33}{42}$	$\frac{72}{42} = 1$

Marginal distribution of x & y :

x	P_{x_i}
0	$18/72$
1	$24/72$
2	$30/72$

y	P_{y_j}
1	$15/72$
2	$24/72$
3	$33/72$

Conditional probability distribution:

i). x given y

$$P_x(x_i/y_j) = \frac{P(x_i; y_j)}{P_y(y_j)}$$

$x=0$ given y

y	$P_x(x_i/y_j)$
1	$3/15 = \frac{1}{5}$
2	$\frac{6}{24} = \frac{1}{4}$
3	$\frac{9}{33} = \frac{3}{11}$

$\sum = 1$

$x=1$ given y

y	$P_x(x_i/y_j)$
1	$\frac{5}{15} = \frac{1}{3}$
2	$\frac{8}{24} = \frac{1}{3}$
3	$\frac{11}{33} = \frac{1}{3}$

$x=2$ given y

y	$P_x(x_i/y_j)$
1	$7/15$
2	$10/24$
3	$13/33$

ii). y given x

$$P_y(y_j/x_i) = \frac{P(x_i; y_j)}{P_x(x_i)}$$

$y=1$ given x

x	$P_y(y_j/x_i)$
0	$\frac{3}{18} = \frac{1}{6}$
1	$\frac{5}{24}$
2	$\frac{7}{30}$

$y=2$ given x

x	$P_y(y_j/x_i)$
0	$6/18 = \frac{1}{3}$
1	$8/24 = \frac{1}{3}$
2	$\frac{10}{30} = \frac{1}{3}$

$y=3$ given x

x	$P_y(y_j/x_i)$
0	$\frac{9}{18} = \frac{1}{2}$
1	$\frac{11}{24}$
2	$\frac{13}{30}$

probability distribution of $(x+y)$:

$x+y$	$P(x+y)$
1	$\frac{3}{72}$
2	$\frac{6}{72} + \frac{5}{72} = \frac{11}{72}$
3	$\frac{9}{72} + \frac{8}{72} + \frac{7}{72} = \frac{24}{72}$
4	$\frac{11}{72} + \frac{10}{72} = \frac{21}{72}$
5	$13/72$

- ③. The joint probability mass fn X & Y is

$x \backslash y$	0	1	2	P_{xi}
0	0.1	0.04	0.02	0.16
1	0.08	0.20	0.06	0.34
2	0.06	0.14	0.30	0.50
P_{yj}	0.24	0.38	0.38	1

Find the marginal probability mass fn X & Y , $P(X \leq 1, Y \leq 1)$ and verify whether X & Y are independent or not.

Sol:

Marginal distribution of X & Y .

x	P_{xi}
0	0.16
1	0.34
2	0.50

y	P_{yj}
0	0.24
1	0.38
2	0.38

$$P(X \leq 1, Y \leq 1) = P(X=0, Y=0) + P(X=0, Y=1) + P(X=1, Y=0) + P(X=1, Y=1)$$

$$= 0.1 + 0.04 + 0.08 + 0.20$$

$$= 0.42$$

since, X and Y are independent.

$$P(x_i, y_j) = P(x_i) \cdot P(y_j)$$

$$\text{let } x_i = 0, y_j = 0$$

$$P(0,0) = P_x(0) \cdot P_y(0)$$

$$0.1 = 0.16 \times 0.24$$

$$0.1 \neq 0.0384$$

$\therefore X$ & Y are not independent.

- ④ 3 balls are drawn at random without replacement from a box containing 2 white, 3 red and 4 black balls. If X denotes the number of white balls drawn and Y denotes the no. of red balls drawn. Find the joint probability distribution of (X, Y) .

Sol:

Given,

$X \rightarrow$ white balls (2)

$Y \rightarrow$ Red balls (3)

$x \backslash y$	0	1	2	3
0	$\frac{1}{21}$	$\frac{3}{14}$	$\frac{1}{7}$	$\frac{1}{84}$
1	$\frac{1}{7}$	$\frac{2}{7}$	$\frac{1}{14}$	0
2	$\frac{1}{21}$	$\frac{1}{38}$	0	0

$$\text{total} = 9$$

$$\text{select} = 3$$

$$P(X=0, Y=0) = \frac{{}^4C_3}{{}^9C_3}$$

$$= \frac{4}{84} = \frac{1}{21}$$

$$P(X=0, Y=1) = \frac{{}^3C_1 \times {}^4C_2}{{}^9C_3}$$

$$= \frac{3 \times 6}{84} = \frac{3}{14}$$

$$P(X=0, Y=2) = \frac{{}^3C_2 \times {}^4C_1}{{}^9C_3}$$

$$= \frac{3 \times 4}{84} = \frac{1}{7}$$

$$P(X=0, Y=3) = \frac{{}^3C_3}{{}^9C_3}$$

$$= \frac{1}{84}$$

$$P(X=1, Y=0) = \frac{{}^2C_1 \times {}^4C_2}{{}^9C_3}$$

$$= \frac{2 \times 6}{84} = \frac{1}{7}$$

$$P(X=1, Y=1) = \frac{{}^2C_1 \times {}^3C_1 \times {}^4C_1}{{}^9C_3}$$

$$= \frac{2 \times 3 \times 4}{84} = \frac{2}{7}$$

$$P(X=1, Y=2) = \frac{{}^2C_1 \times {}^3C_2}{{}^9C_3}$$

$$= \frac{2 \times 3}{84} = \frac{1}{14}$$

$$P(X=1, Y=3) = \frac{{}^2C_1 \times {}^3C_3}{{}^9C_3}$$

$= 0$ [select three balls only]

$$P(X=2, Y=0) = \frac{{}^2C_2 \times {}^4C_1}{{}^9C_3}$$

$$= \frac{1 \times 4}{84} = \frac{1}{21}$$

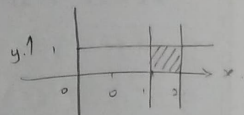
$$P(X=2, Y=1) = \frac{{}^2C_2 \times {}^3C_1}{{}^9C_3}$$

$$= \frac{1 \times 3}{84} = \frac{1}{28}$$

$$P(X=2, Y=2) \text{ \& } P(X=2, Y=3) = 0$$

15. The joint pdf of a two dimensional RV (x, y) is given by $f(x, y) = xy^2 + \frac{x^2}{8}$, $0 \leq x \leq 2$, $0 \leq y \leq 1$. compute $P(X > 1)$, $P(Y < 1/2)$, $P(X > 1 | Y < 1/2)$, $P(Y < 1/2 | X > 1)$, $P(X < Y)$ and $P(X + Y \leq 1)$

sol:



$$i). P(X > 1) = \int_{y=0}^1 \int_{x=1}^2 f(x, y) dx dy.$$

$$= \int_0^1 \left[\int_1^2 xy^2 + \frac{x^2}{8} dx \right] dy.$$

$$= \int_0^1 \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_1^2 dy.$$

$$= \int_0^1 \left(\frac{4y^2}{2} + \frac{8}{24} - \frac{y^2}{2} - \frac{1}{24} \right) dy$$

$$= \left[\frac{3y^3}{6} + \frac{7y}{24} \right]_0^1$$

$$= \frac{1}{2} + \frac{7}{24}$$

$$= \frac{19}{24}$$

$$ii). P(Y < 1/2) = \int_{y=0}^{1/2} \int_{x=0}^2 f(x, y) dx dy.$$

$$= \int_0^{1/2} \left[\int_0^2 xy^2 + \frac{x^2}{8} dx \right] dy$$

$$= \int_0^{1/2} \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_0^2 dy.$$

$$= \int_0^{1/2} \left(\frac{4y^2}{2} + \frac{8}{24} \right) dy.$$

$$= \int_0^{1/2} \left(2y^2 + \frac{1}{3} \right) dy.$$

$$= \left[\frac{2y^3}{3} + \frac{y}{3} \right]_0^{1/2}$$

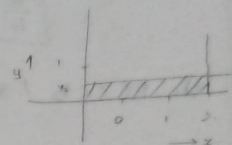
$$= \frac{2(1/8)}{3} + \frac{1}{6} - 0.$$

$$= \frac{1}{12} + \frac{1}{6}$$

$$= \frac{1+2}{12}$$

$$= \frac{3}{12}$$

$$= \frac{1}{4}$$



$$iii). P(X > 1 | Y < 1/2) = \frac{P(X > 1; Y < 1/2)}{P(Y < 1/2)}$$

$$= \frac{\int_{y=0}^{1/2} \int_{x=1}^2 f(x, y) dx dy}{1/4}$$

$$= 4 \int_0^{1/2} \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_1^2 dy.$$

$$= 4 \int_0^{1/2} \left[\frac{4y^2}{2} + \frac{8}{24} - \frac{y^2}{2} - \frac{1}{24} \right] dy.$$

$$= 4 \left[\frac{3y^3}{6} + \frac{7y}{24} \right]_0^{1/2}$$

$$= 4 \left[\frac{1}{16} + \frac{7}{48} \right]$$

$$= 4 \left[\frac{10}{48} \right]$$

$$= \frac{5}{6}$$

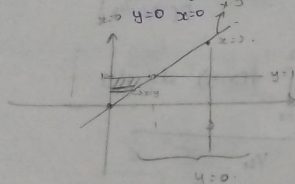
$$\text{iv). } P(Y < 1/2 \mid X > 1) = \frac{P(Y < 1/2, X > 1)}{P(X > 1)}$$

$$= \frac{\int_0^{1/2} \int_1^2 f(x, y) dx dy}{19/24}$$

$$= \frac{10/5 \times 24}{48 \times 19}$$

$$= \frac{5}{19}$$

$$\text{v). } P(X < Y) = \int_0^1 \int_0^y f(x, y) dx dy$$



$$= \int_0^1 \left[\int_0^y xy^2 + \frac{x^2}{8} dx \right] dy$$

$$= \int_0^1 \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_0^y dy$$

$$= \int_0^1 \frac{y^4}{2} + \frac{y^3}{24} dy$$

$$= \left[\frac{y^5}{10} + \frac{y^4}{24 \times 4} \right]_0^1$$

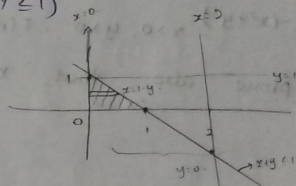
$$= \frac{1}{10} + \frac{1}{96}$$

$$= \frac{96 + 10}{960}$$

$$= \frac{106}{960}$$

$$= \frac{53}{480}$$

$$\text{vi). } P(X+Y \leq 1)$$



$$= \int_0^1 \int_0^{1-y} f(x, y) dx dy$$

$$= \int_0^1 \left[\int_0^{1-y} xy^2 + \frac{x^2}{8} dx \right] dy$$

$$= \int_0^1 \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_0^{1-y} dy$$

$$= \int_0^1 \left[\frac{(1-y)^2 y^2}{2} + \frac{(1-y)^3}{24} \right] dy$$

$$= \int_0^1 \left[\frac{y^2 + y^4 - 2y^3}{2} + \frac{(1-y)^3}{24} \right] dy$$

$$= \left[\frac{1}{2} \left(\frac{y^3}{3} - \frac{2y^4}{4} + \frac{y^5}{5} \right) + \frac{(1-y)^4}{(-1) \times 24} \right]_0^1$$

$$= \left[\frac{1}{2} \left(\frac{1}{3} - \frac{2}{4} + \frac{1}{5} \right) + 0 - \frac{1}{2}(0) + \frac{1}{4 \times 24} \right]$$

$$= \left[\frac{1}{2} \left(\frac{10-15+6}{30} \right) + \frac{1}{96} \right]$$

$$= \frac{1}{60} + \frac{1}{96}$$

$$= \frac{13}{480}$$

⑥. The joint pdf of the RV (x, y) is given by $f(x, y) = kxy e^{-(x^2+y^2)}$ $x > 0, y > 0$. Find the value of k and prove also that x and y are independent.

⑦. Given $f_{xy}(x, y) = cx(x-y)$, $0 < x < 2$, $-x < y < x$ and 0 elsewhere, a) evaluate c

b) find $f_x(x)$

c) $f_{y/x}(y/x)$

d) $f_y(y)$

⑧. Sol: Given, $f(x, y) = kxy e^{-(x^2+y^2)}$, $x > 0, y > 0$.

$$\int_{y=0}^{\infty} \int_{x=0}^{\infty} f(x, y) dx dy = 1.$$

$$\int_{y=0}^{\infty} \int_{x=0}^{\infty} kxy e^{-x^2-y^2} dx dy = 1.$$

$$k \left[\int_{x=0}^{\infty} x e^{-x^2} dx \cdot \int_{y=0}^{\infty} y e^{-y^2} dy \right] = 1.$$

$$\text{Let } t = x^2$$

$$dt = 2x dx.$$

$$\frac{dt}{2} = x dx.$$

Similarly for y .

$$x=0 \rightarrow t=0$$

$$x=\infty \rightarrow t=\infty.$$

$$k \left[\int_0^{\infty} \frac{e^{-t}}{2} dt \cdot \int_0^{\infty} \frac{e^{-t}}{2} dt \right] = 0.$$

$$\frac{k}{4} \left[\left(\frac{e^{-t}}{-1} \right)_0^{\infty} \cdot \left(\frac{e^{-t}}{-1} \right)_0^{\infty} \right] = 0.$$

$$\frac{k}{4} \left[(0+1)(0+1) \right] = 0.$$

$$\frac{k}{4} (1) = 0.$$

$$\boxed{k=4}$$

$$f(x, y) = 4xy e^{-(x^2+y^2)}, x > 0, y > 0.$$

⑨.

x and y are independent.

$$f(x, y) = f(x) f(y)$$

$$f_{xy}(x, y) = f_x(x) \cdot f_y(y)$$

$$f_x(x) = \int_{y=0}^{\infty} f(x, y) dy.$$

$$= \int_0^{\infty} 4xy e^{-x^2-y^2} dy.$$

$$= 4x e^{-x^2} \int_0^{\infty} y e^{-y^2} dy.$$

let $t = y^2$.

$$\frac{dt}{2} = y dy$$

$$y=0 \rightarrow t=0.$$

$$y=\infty \rightarrow t=\infty.$$

$$= 4x e^{-x^2} \int_0^{\infty} \frac{e^{-t}}{2} dt.$$

$$= 2x e^{-x^2} \left[\frac{e^{-t}}{-1} \right]_0^{\infty}$$

$$= 2x e^{-x^2} (0+1)$$

$$f_x(x) = 2x e^{-x^2}, \quad x > 0.$$

iii) dy.

$$f_y(y) = 2y e^{-y^2}, \quad y > 0$$

$$f_x(x) \cdot f_y(y) = (2x e^{-x^2}) \cdot (2y e^{-y^2}) \\ = 4xy e^{-x^2-y^2} \rightarrow \textcircled{2}.$$

from ① & ②.

$$f(x,y) = f_x(x) \cdot f_y(y)$$

$\therefore x$ and y are independent.

Hence proved.

Sol

Given, $f_{xy}(x,y) = c x(x-y)$, $0 < x < 2$,
 $-x < y < x$.

a). to find c .

$$\int_{y=-x}^x \int_{x=0}^2 f(x,y) dx dy = 1.$$

$$\left[\int_{-x}^x \int_0^2 c x(x-y) dx \right] dy = 1.$$

$$\int_{-x}^x \left[\int_0^2 c(x^2 - xy) dx \right] dy = 1.$$

$$\int_{-x}^x \left[c \left(\frac{x^3}{3} - \frac{xy^2}{2} \right) \right] dy = 1.$$

$$c \left[\frac{x^3}{3} - \frac{xy^2}{2} \right]_{-x}^x$$

$$c \int_0^2 \left[\int_{-x}^x x^2 - xy dy \right] dx = 1.$$

$$c \int_0^2 \left[x^2 y - \frac{xy^2}{2} \right]_{-x}^x dx = 1.$$

$x^2 \rightarrow$ even, $-xy \rightarrow$ odd.

$$c \int_0^2 \left[x^3 - \frac{x^3}{2} + x^3 + \frac{x^3}{2} \right] dx = 1.$$

$$c \int_0^2 2x^3 dx = 1.$$

$$c \left[\frac{2x^4}{4} \right]_0^2 = 1$$

$$c \left(\frac{16}{2} \right) = 1.$$

$$c = \frac{1}{8}.$$

$$\begin{aligned}
 b). \quad f_x(x) &= \int_{-x}^x \frac{1}{8} (x^2 - xy) dy \\
 &= \frac{2}{8} \int_0^x x^2 dy \\
 &= \frac{1}{4} [x^2 y]_0^x \\
 &= \frac{1}{4} (x^3)
 \end{aligned}$$

$$f_x(x) = \frac{x^3}{4}, \quad 0 < x < 2.$$

$$\begin{aligned}
 c). \quad f_{y|x}(y/x) &= \frac{f_{xy}(x,y)}{f_x(x)} \\
 &= \frac{\frac{1}{8} (x^2 - xy)}{\frac{x^3}{4}} = \frac{x(x-y)}{2x^3}
 \end{aligned}$$

$$f_{y|x}(y/x) = \frac{x-y}{2x^2}, \quad 0 < x < 2, \quad -x < y < x.$$

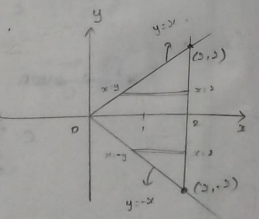
$$d). \quad f_y(y) = \int_{x=0}^2 \frac{1}{8} (x^2 - xy) dx.$$

$$= \frac{1}{8} \int_0^2 \left[\frac{x^3}{3} - \frac{x^2 y}{2} \right]_0^2$$

$$= \frac{1}{8} \left[\frac{8}{3} - \frac{4y}{2} \right]$$

$$= \frac{1}{3} - \frac{y}{4}.$$

$$f_y(y) = \frac{4-3y}{12}, \quad -x < y < x.$$



Correlation Coefficient:

$$r_{xy} = \frac{E\{[X-E(X)][Y-E(Y)]\}}{\sqrt{E\{[X-E(X)]^2\} \cdot E\{[Y-E(Y)]^2\}}}$$

$$r_{xy} = \frac{\text{cov}(X,Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}$$

$$r_{xy} = \frac{E(XY) - E(X)E(Y)}{\sqrt{[E(X^2) - E^2(X)][E(Y^2) - E^2(Y)]}}$$

continuous

$$r_{xy} = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

discrete

$$\therefore \text{cov}(X,Y) = E(XY) - E(X)E(Y)$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2$$

properties:

$$1. \quad -1 \leq r_{xy} \leq 1$$

2. Correlation coefficient is independent of change of origin & scale.

3. Two independent RV's X & Y are uncorrelated.

8. If x and y are RV having the joint density function $f_{xy}(x,y) = \begin{cases} \frac{1}{8}(6-x-y), & 0 < x < 2, \\ & 2 < y < 4 \\ 0, & \text{otherwise} \end{cases}$

Find i). $P(x+y < 3)$, ii). $P(x < 1, y < 3)$, iii). $P(x < 1 | y < 3)$

9. If the joint probability function of RV (x,y)

$$f_{xy}(x,y) = \begin{cases} 8xy, & 0 < x < 1, 0 < y < x \\ 0, & \text{otherwise} \end{cases}$$

Find i) $P(y < 1/8 | x < 1/2)$ ii) $f_y(y|x)$

10. If joint density function of x and y is given by $f_{xy}(x,y) = \begin{cases} e^{-(x+y)}, & x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$

Find i) Marginal density functions of x & y and verify x and y are independent or not

ii). $P(x < 1)$

iii). $P(x+y < 1)$

iv). $P(0 < x < 1 | y \neq 2)$

v). $P(x > y)$

11. Given the joint probability density function of (x,y) as $f_{xy}(x,y) = \begin{cases} 8xy, & 0 < x < y < 1 \\ 0, & \text{otherwise} \end{cases}$. Find

i) Marginal and conditional probability functions of x and y .

ii). Are x and y independent?

iii). Find $P(x+y > 1)$

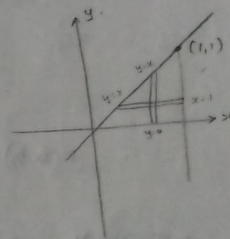
$$f_{xy}(x,y) = 8xy, 0 < x < 1, 0 < y < x$$

$$f_x(x) = \int_{y=0}^x f(x,y) dy = \int_0^x 8xy dy$$

$$= \left[8x \frac{y^2}{2} \right]_0^x = 4x^3$$

$$f_y(y) = \int_{x=y}^1 f(x,y) dx = \int_y^1 8xy dx$$

$$= \left[\frac{8x^2}{2} y \right]_y^1 = 4y - 4y^3$$



i). $P(y < 1/8 | x < 1/2) = \frac{P(x < 1/2, y < 1/8)}{P(x < 1/2)}$

$$P(x < 1/2, y < 1/8) = \int_{y=0}^{1/8} \int_{x=y}^{1/2} f(x,y) dx dy$$

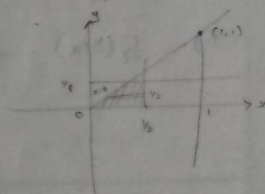
$$= \int_0^{1/8} \left[\frac{8x^2}{2} y \right]_y^{1/2} dy$$

$$= \int_0^{1/8} (y - 4y^3) dy$$

$$= \left[\frac{y^2}{2} - \frac{4y^4}{4} \right]_0^{1/8}$$

$$= \frac{1}{128} - \frac{1}{1024}$$

$$= \frac{31}{1024}$$



For $f_x(x)$ & $f_y(y)$
take integration limit
for others take
entire limit

ii) $f_Y(y/x)$

$$P(X < \frac{1}{2}) = \int_{x=0}^{1/2} f_X(x) dx$$

$$= \int_0^{1/2} 4x^3 dx$$

$$= \left[\frac{4x^4}{4} \right]_0^{1/2}$$

$$= \frac{1}{16} - 0^4$$

$$P(X < 1/2) = \frac{1}{16}$$

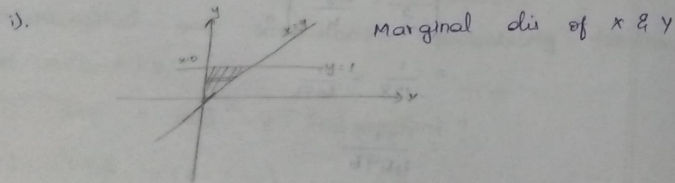
$$P(Y < \frac{1}{8} | X < \frac{1}{2}) = \frac{31}{4096} \times \frac{16}{1}$$

$$= \frac{31}{256}$$

ii). $f_Y(y/x) = \frac{f_{XY}}{f_X(x)} = \frac{8xy}{4x^3}$

$$= \frac{2y}{x^2}$$

11) Sol: Given, $f(x,y) = \begin{cases} 8xy, & 0 < x < y < 1 \\ 0, & \text{otherwise} \end{cases}$



$$f_X(x) = \int_x^1 8xy dy$$

$$= \left[\frac{8xy^2}{2} \right]_x^1$$

$$f_X(x) = 4x - 4x^3, \quad 0 < x < 1$$

$$f_Y(y) = \int_{x=0}^y 8xy dx$$

$$= \left[\frac{8x^2y}{2} \right]_0^y$$

$$f_Y(y) = 4y^3, \quad 0 < y < 1$$

Conditional Probability of X given Y & Y given X .

$$f_{X/Y}(x/y) = \frac{f(x,y)}{f_Y(y)} = \frac{8xy}{4y^3}$$

$$= \frac{2x}{y^2}, \quad 0 < x < y < 1$$

$$f_{Y/X}(y/x) = \frac{f(x,y)}{f_X(x)}$$

$$= \frac{8xy}{4x - 4x^3} = \frac{8xy}{4x(1-x^2)}$$

$$= \frac{2y}{(1-x^2)}, \quad 0 < x < y < 1$$

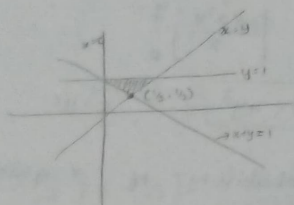
ii) $f_{xy}(x,y) = f_x(x) \cdot f_y(y)$

$$8xy = (4x - 4x^3)(4y^3)$$

$$8xy \neq 16xy^3 - 16x^3y^3$$

$\therefore x$ & y are not independent.

iii) $P(x+y > 1)$



$$= \int_{y=1/2}^1 \int_{x=1-y}^1 f(x,y) dx dy$$

$$= \int_{1/2}^1 \left[\frac{8x^2y}{2} \right]_{1-y}^1 dy$$

$$= \int_{1/2}^1 4y^3 - 4y(1-y)^2 dy$$

$$= \int_{1/2}^1 4y^3 - 4y + 8y^2 dy$$

$$= \int_{1/2}^1 8y^2 - 4y dy$$

$$= \left[\frac{8y^3}{3} - \frac{4y^2}{2} \right]_{1/2}^1$$

$$= \frac{8}{3} - 2(1) - \frac{8}{3} + 4\left(\frac{1}{4}\right)$$

$$= \frac{8}{3} - 2 - \frac{1}{3} + \frac{1}{2}$$

$$= \frac{7}{3} + \frac{-4+1}{2}$$

$$= \frac{7}{3} - \frac{3}{2}$$

$$= \frac{5}{6}$$

⑦ Two RV x & y have joint density fn

$$f(x,y) = \begin{cases} 2-x-y, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find

covariance and correlation coefficient of x & y

⑧ sol: Given, $f_{xy}(x,y) = \begin{cases} \frac{1}{2}(6-x-y), & 0 < x < 2, 2 < y < 4 \\ 0, & \text{otherwise} \end{cases}$

⑩ sol: Given, $f(x,y) = e^{-(x+y)}, x \geq 0, y \geq 0$

i) $f_x(x) = \int_0^{\infty} f(x,y) dy$

$$= \int_0^{\infty} e^{-x-y} dy$$

$$= e^{-x} \int_0^{\infty} e^{-y} dy$$

$$= e^{-x} \left[\frac{e^{-y}}{-1} \right]_0^{\infty} = e^{-x} [0+1]$$

$$\boxed{f_x(x) = e^{-x}}$$

$$f_Y(y) = e^y \int_0^\infty e^{-x-y} dx$$

$$= e^y \left[\frac{e^{-x}}{-1} \right]_0^\infty$$

$$= e^y (0-1)$$

$$\boxed{= e^{-y}}$$

Marginal density ~~fn~~ of x & y are

$$f_X(x) = e^{-x}, \quad f_Y(y) = e^{-y}$$

Condition for independent:

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y)$$

$$e^{-x-y} = e^{-x} \cdot e^{-y}$$

$$e^{-(x+y)} = e^{-(x+y)}$$

$\therefore x$ and y are independent.

ii) $P(X < 1)$

$$= \int_0^1 f_X(x) dx$$

$$= \int_0^1 e^{-x} dx = \left[\frac{e^{-x}}{-1} \right]_0^1$$

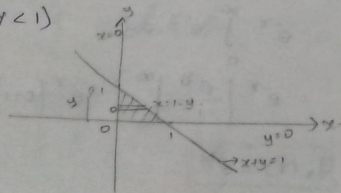
$$= -e^{-1} + 1$$

$$= 1 - e^{-1}$$

$$= 1 - 0.3678$$

$$\boxed{= 0.6321}$$

iii) $P(X+Y < 1)$



$$= \int_{y=0}^1 \int_{x=0}^{1-y} e^{-x-y} dx dy$$

$$= \int_{y=0}^1 e^{-y} \left[\frac{e^{-x}}{-1} \right]_0^{1-y} dy$$

$$= \int_0^1 e^{-y} [-e^{-(1-y)} + 1] dy$$

$$= \int_0^1 e^{-y} [-e^{-1+y} + 1] dy$$

$$= \left[\frac{e^{-y}}{-1} + e^{-y} \right]_0^1$$

$$= \left[\frac{e^{-1}}{-1} - e^{-1} + 1 \right]$$

$$= 1 - 2e^{-1}$$

$$\boxed{= 0.2642}$$

iv) $P(0 < X < 1 \mid Y \neq 0) = P(0 < X < 1)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) P(B)}{P(B)} = P(A)$$

$$P(0 < X < 1) = \int_0^1 f_X(x) dx$$

$$= \int_0^1 e^{-x} dx$$

$$= \left[\frac{e^{-x}}{-1} \right]_0^1$$

$$= 1 - e^{-1}$$

$$\boxed{= 0.6321}$$

[$P(A \cap B) = P(A) P(B)$
if A & B are
independent]

v). $P(x > y)$

$$= \int_{x=0}^{\infty} \int_{y=0}^x e^{-x} e^{-y} dy dx.$$

$$= \int_0^{\infty} e^{-x} \left[\frac{e^{-y}}{-1} \right]_0^x dx.$$

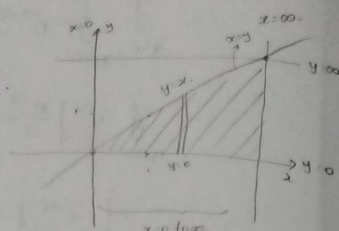
$$= \int_0^{\infty} e^{-x} (-e^{-x} + 1) dx.$$

$$= \int_0^{\infty} -e^{-2x} + e^{-x} dx.$$

$$= \left[-\frac{e^{-2x}}{(-2)} + \frac{e^{-x}}{(-1)} \right]_0^{\infty}$$

$$= \left[0 - 0 - \frac{1}{2} + 1 \right]$$

$$= \frac{1}{2}$$



$$= \frac{1}{8} \int_2^3 \left[18 - \frac{9}{2} - 6y + \frac{6y^2}{2} - 3y - \frac{y^3}{2} + y^3 \right] dy.$$

$$= \frac{1}{8} \int_2^3 \left[\frac{27}{2} - 6y + \frac{y^2}{2} \right] dy.$$

$$= \frac{1}{8} \left[\frac{27}{2} y - \frac{6y^2}{2} + \frac{y^3}{6} \right]_2^3$$

$$= \frac{1}{8} \left[\frac{81}{2} - \frac{54}{2} + \frac{27}{6} - 27 + 12 - \frac{4}{3} \right]$$

$$= \frac{1}{8} \left[\frac{27}{2} + \frac{27}{6} - 15 - \frac{4}{3} \right]$$

$$= \frac{1}{8} \left[\frac{81 + 27 - 90 - 8}{6} \right]$$

$$= \frac{1}{8} \left[\frac{10}{6} \right]$$

$$= \frac{5}{24}$$

ii). $P(x < 1, y < 3) = \int_{x=0}^1 \int_{y=0}^3 \frac{1}{8} (6-x-y) dy dx.$

$$= \frac{1}{8} \int_0^1 \left[6y - xy - \frac{y^2}{2} \right]_0^3 dx.$$

$$= \frac{1}{8} \int_0^1 \left[18 - 3x - \frac{9}{2} - 12 + 2x + \frac{9}{2} \right] dx.$$

$$= \frac{1}{8} \int_0^1 \left(8 - x - \frac{9}{2} \right) dx$$

$$= \frac{1}{8} \left[8x - \frac{x^2}{2} - \frac{9x}{2} \right]_0^1$$

$$= \frac{1}{8} \left[8 - \frac{1}{2} - \frac{9}{2} \right] = \frac{1}{8} \left[8 - \frac{10}{2} \right]$$

$$= \frac{1}{8} \left[\frac{6}{2} \right]$$

$$= \frac{3}{8}$$

2) sol:-

Given. $f(x, y) = \begin{cases} \frac{1}{8} (6-x-y), & 0 \leq x < 2, 0 \leq y < 4 \\ 0, & \text{otherwise} \end{cases}$

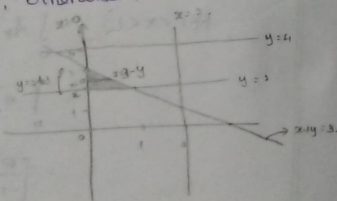
i). $P(x+y < 3)$

$$= \int_{y=0}^3 \int_{x=0}^{3-y} \frac{1}{8} (6-x-y) dx dy$$

$$= \frac{1}{8} \int_0^3 \left[6x - \frac{x^2}{2} - xy \right]_0^{3-y} dy$$

$$= \frac{1}{8} \int_0^3 \left[6(3-y) - \frac{(3-y)^2}{2} - (3-y)y \right] dy$$

$$= \frac{1}{8} \int_0^3 \left[18 - 6y - \left(\frac{9+y^2-6y}{2} \right) - 3y + y^2 \right] dy$$



$$\text{iii). } P(X < 1 | Y < 3) = \frac{P(X < 1, Y < 3)}{P(Y < 3)}$$

$$P(X < 1, Y < 3) = \frac{3}{8}$$

$$\begin{aligned} P(Y < 3) &= \int_{y=2}^3 \int_{x=0}^2 \frac{1}{8} (6-x-y) dx dy \\ &= \frac{1}{8} \int_2^3 \left(6x - \frac{x^2}{2} - xy \right)_0^2 dy \\ &= \frac{1}{8} \int_2^3 (12 - 2 - 2y) dy \\ &= \frac{1}{8} \left[10y - \frac{2y^2}{2} \right]_2^3 \\ &= \frac{1}{8} [30 - 9 - 20 + 4] \\ &= \frac{1}{8} (34 - 29) \\ &= \frac{5}{8} \end{aligned}$$

$$\therefore P(X < 1 | Y < 3) = \frac{\frac{3}{8}}{\frac{5}{8}} = \frac{3}{5}$$

(b) Sol:
Given, $f(x, y) = \begin{cases} 2-x-y, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$

$$f_X(x) = \int_0^1 f(x, y) dy$$

$$= \int_0^1 (2-x-y) dy$$

$$= \left[2y - xy - \frac{y^2}{2} \right]_0^1$$

$$= 2 - x - \frac{1}{2}$$

$$\boxed{f_X(x) = \frac{3}{2} - x, \quad 0 < x < 1}$$

$$f_Y(y) = \int_{x=0}^1 (2-x-y) dx$$

$$= \left(2x - \frac{x^2}{2} - xy \right)_0^1$$

$$= 2 - \frac{1}{2} - y$$

$$\boxed{f_Y(y) = \frac{3}{2} - y, \quad 0 < y < 1}$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$E(X) = \int_{x=0}^1 x f_X(x) dx$$

$$= \int_0^1 x \left(\frac{3}{2} - x \right) dx$$

$$= \int_0^1 \left(\frac{3}{2}x - x^2 \right) dx$$

$$= \left[\frac{3}{2} \cdot \frac{x^2}{2} - \frac{x^3}{3} \right]_0^1$$

$$= \frac{3}{4} - \frac{1}{3}$$

$$\boxed{E(X) = \frac{5}{12}}$$

Similarly

$$E(Y) = \frac{5}{12}$$

$$\begin{aligned}
 E(xy) &= \int_{y=0}^1 \int_{x=0}^y xy f(x,y) dx dy \\
 &= \int_0^1 \int_0^y xy (3-x-y) dx dy \\
 &= \int_0^1 \int_0^y (3xy - x^2y - y^2x) dx dy \\
 &= \int_0^1 \left[\frac{3x^2y}{2} - \frac{x^3y}{3} - \frac{y^2x^2}{2} \right]_0^y dy \\
 &= \int_0^1 \left(\frac{3y^3}{2} - \frac{y^4}{3} - \frac{y^3}{2} \right) dy \\
 &= \int_0^1 \frac{2y^3}{3} dy \\
 &= \left[\frac{2}{3} \cdot \frac{y^4}{4} - \frac{y^3}{6} \right]_0^1 \\
 &= \frac{1}{3} - \frac{1}{6} \\
 &= \boxed{\frac{1}{6}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Cov}(X, Y) &= E(XY) - E(X)E(Y) \\
 &= \frac{1}{6} - \frac{5}{12} \left(\frac{5}{12} \right) \\
 &= \frac{1}{6} - \frac{25}{144} \\
 &= \frac{144 - 150}{864} \\
 &= \frac{-6}{864} \\
 &= \frac{-1}{144}
 \end{aligned}$$

Correlation coefficient:

$$r_{xy} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_{x=0}^1 x^2 f_X(x) dx$$

$$= \int_0^1 x^2 \left(\frac{3}{2} - x \right) dx$$

$$= \int_0^1 \left(\frac{3}{2}x^2 - x^3 \right) dx$$

$$= \left[\frac{3}{2} \cdot \frac{x^3}{3} - \frac{x^4}{4} \right]_0^1$$

$$= \frac{1}{2} - \frac{1}{4}$$

$$= \boxed{\frac{1}{4}}$$

$$\text{Similarly } E(Y^2) = \frac{1}{4}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \frac{1}{4} - \frac{25}{144}$$

$$= \frac{144 - 100}{576}$$

$$= \frac{44}{576}$$

$$= \frac{11}{144}$$

$$= \boxed{\frac{11}{144}}$$

$$\text{Similarly } \text{Var}(Y) = \frac{11}{144}$$

$$\begin{aligned} \gamma_{xy} &= \frac{-1}{144} \\ &= \frac{-1/144}{\sqrt{\frac{11}{144}} \sqrt{\frac{11}{144}}} \\ &= \frac{-1/144}{\frac{11}{144}} \\ &= \frac{-1}{11} \end{aligned}$$

$$\gamma_{xy} = -0.0909$$

- (13) The joint pdf of the random variable x and y is defined as follows

$$f_{xy}(x, y) = \begin{cases} 25e^{-5y}, & 0 < x < 1, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

what is covariance of x and y

- (14) The joint probability density functions of x and y is $f_{xy}(x, y) = \begin{cases} x-y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$

obtain the correlation coefficient between x & y .

- (15) The joint probability mass function of x and y is given below.

$y \backslash x$	-1	1
0	$\frac{1}{8}$	$\frac{3}{8}$
1	$\frac{2}{8}$	$\frac{2}{8}$

Find the correlation between x & y .

$y \backslash x$	-1	1	P_y
0	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{4}{8}$
1	$\frac{2}{8}$	$\frac{2}{8}$	$\frac{4}{8}$
P_x	$\frac{3}{8}$	$\frac{5}{8}$	1

marginal distribution of x & y .

x	$P_x(x)$	y	$P_y(y)$
-1	$\frac{3}{8}$	0	$\frac{4}{8}$
1	$\frac{5}{8}$	1	$\frac{4}{8}$

$$\begin{aligned} E(x) &= \sum x P_x(x) \\ &= -\frac{3}{8} + \frac{5}{8} \\ &= \frac{2}{8} = \frac{1}{4} \end{aligned}$$

$$\begin{aligned} E(y) &= \sum y P_y(y) \\ &= \frac{4}{8}(0) + \frac{4}{8}(1) \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} E(x^2) &= \sum x^2 P_x(x) \\ &= (-1)^2 \frac{3}{8} + (1)^2 \frac{5}{8} \\ &= \frac{8}{8} = 1 \end{aligned}$$

$$E(Y^2) = \sum y^2 P_Y(y)$$

$$= \frac{4}{8}(1)^2 + \frac{4}{8}(0)^2$$

$$= \frac{1}{2}$$

$$E(XY) = \sum xy P_{XY}(x, y)$$

$$= (1)(-1) \frac{2}{8} + (1)(1) \frac{2}{8}$$

$$= -\frac{2}{8} + \frac{2}{8}$$

$$= 0$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$= 0 - \frac{1}{4} \times \frac{1}{2}$$

$$= -\frac{1}{8}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$= 1 - \left(\frac{1}{4}\right)^2$$

$$= \frac{15}{16}$$

$$\text{Var}(Y) = E(Y^2) - (E(Y))^2$$

$$= \frac{1}{2} - \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{2} - \frac{1}{4}$$

$$= \frac{1}{4}$$

Correlation coefficient,

$$r_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}$$

$$= \frac{-1/8}{\sqrt{\frac{15}{16}} \sqrt{\frac{1}{4}}}$$

$$= \frac{-1/8}{\frac{\sqrt{15}}{4} \cdot \frac{1}{2}} = \frac{-1/8}{\frac{\sqrt{15}}{8}} = \frac{-1}{\sqrt{15}}$$

$$= -\frac{1}{\sqrt{15}} = -0.2582$$

Note:

$$* r_{XY} = 0$$

(i.e.) when $\text{Cov}(X, Y) = 0$

$\therefore X$ and Y are uncorrelated.

$$* r_{XY} \neq 0$$

(i.e.) when $\text{Cov}(X, Y) \neq 0$

$\therefore X$ and Y are correlated.

(B) Sol:

$$\text{Given, } f_{XY}(x, y) = \begin{cases} 25e^{-5y}, & 0 < x < 2, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \int_0^{\infty} \int_0^2 x \cdot f(x, y) dx dy$$

$$= \int_0^{\infty} 25e^{-5y} \left[\frac{x^2}{2} \right]_0^2 dy$$

$$= \int_0^{\infty} 25e^{-5y} \times (0-0) dy$$

$$= 50 \left[\frac{e^{-5y}}{-5} \right]_0^{\infty}$$

$$= 10(0+1)$$

$$\boxed{E(x) = 10}$$

$$E(y) = \int_0^{\infty} \int_0^2 y \cdot 25e^{-5y} dx dy$$

$$= \int_0^{\infty} 25y e^{-5y} \left[x \right]_0^2 dy$$

$$= 50 \int_0^{\infty} y e^{-5y} dy$$

$$\text{let } u = y, \quad v = \frac{e^{-5y}}{-5}$$

$$du = 1, \quad v_1 = \frac{e^{-5y}}{25}$$

$$= 50 \left[(y) \frac{e^{-5y}}{-5} - \frac{e^{-5y}}{25} \right]_0^{\infty}$$

$$= 50 \left(0 - 0 - 0 + \frac{1}{25} \right)$$

$$= \frac{50}{25}$$

$$E(y) = \boxed{2}$$

$$E(xy) = \int_0^{\infty} \int_0^2 xy \cdot 25e^{-5y} dx dy$$

$$= \int_0^{\infty} y \cdot 25e^{-5y} \left[\frac{x^2}{2} \right]_0^2 dy$$

$$= \int_0^{\infty} 50y e^{-5y} dy = 2$$

$$\frac{x^2}{2} = \frac{4}{2} = 2$$

$$COV = E(xy) - E(x)E(y)$$

$$= 2 - (10)(2)$$

$$= 2 - 20$$

$$COV = -18$$

after the process of Normalization.

$\div 10$,

$$COV = \frac{2}{10} - \frac{10}{10} \cdot \frac{2}{10}$$

$$= 0.2 - (1)(0.2)$$

$$\boxed{COV = 0}$$

(19)

Sol: Given, $f_{xy}(x,y) = \begin{cases} x+y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$

$$f_x(x) = \int_0^1 f(x,y) dy$$

$$= \int_0^1 x+y dy$$

$$= \left[xy + \frac{y^2}{2} \right]_0^1$$

$$f_x(x) = x + \frac{1}{2}$$

$$f_y(y) = \int_0^1 x+y dx$$

$$= \left[\frac{x^2}{2} + xy \right]_0^1$$

$$f_y(y) = \frac{1}{2} + y$$

$$E(x) = \int_0^1 x \cdot f_x(x) dx$$

$$= \int_0^1 x^2 + \frac{x}{2} dx$$

$$= \left[\frac{x^3}{3} + \frac{x^2}{4} \right]_0^1$$

$$= \frac{1}{3} + \frac{1}{12}$$

$$E(x) = \frac{7}{12}$$

$$\text{Hly } E(y) = \frac{7}{12}$$

$$E(x, y) = \int_0^1 \int_0^1 xy \cdot f(x, y) \, dx \, dy$$

$$= \int_0^1 \int_0^1 xy^3 + xy^2 \, dx \, dy$$

$$= \int_0^1 \left[\frac{y^3 x^2}{2} + \frac{y^2 x^2}{2} \right]_0^1 dy$$

$$= \int_0^1 \left(\frac{y^3}{2} + \frac{y^2}{2} \right) dy$$

$$= \left[\frac{y^4}{8} + \frac{y^3}{6} \right]_0^1$$

$$= \frac{1}{8} + \frac{1}{6}$$

$$= \frac{2}{24} = \frac{1}{12}$$

$$\text{Cov}(x, y) = E(xy) - E(x)E(y)$$

$$= \frac{1}{12} - \frac{7}{12} \left(\frac{7}{12} \right)$$

$$= \frac{1}{12} - \frac{49}{144}$$

$$= \frac{144 - 49}{144}$$

$$= \frac{-35}{144}$$

$$= \frac{-1}{4.14}$$

Correlation coefficient,

$$r_{xy} = \frac{\text{Cov}(x, y)}{\sqrt{\text{Var}(x)} \sqrt{\text{Var}(y)}}$$

$$\text{Var}(x) = E(x^2) - (E(x))^2$$

$$E(x^2) = \int_0^1 x^2 \cdot f(x) \, dx$$

$$= \int_0^1 x^3 + \frac{x^2}{2} \, dx$$

$$= \left[\frac{x^4}{4} + \frac{x^3}{6} \right]_0^1$$

$$= \frac{1}{4} + \frac{1}{6}$$

$$= \frac{10}{24}$$

$$= \frac{5}{12}$$

$$\text{Hly } E(y^2) = \frac{5}{12}$$

$$\text{Var}(x) = \frac{5}{12} - \frac{49}{144}$$

$$= \frac{720 - 498}{1728}$$

$$= \frac{132}{1728}$$

$$= \frac{11}{144}$$

$$\text{Hly } \text{Var}(y) = \frac{11}{144}$$

$$r_{xy} = \frac{-1/4.14}{\sqrt{\frac{11}{144}} \sqrt{\frac{11}{144}}}$$

$$= \frac{-1/11}{11/11} = -\frac{1}{11}$$

$$r_{xy} = -0.09090$$

10. The following table gives the joint probability distribution of two random variables X and Y . Find $E(X)$, $E(Y)$ and $E(XY)$. Verify whether X and Y are correlated.

$Y \backslash X$	0	1	2	3	$P_Y(y)$
2	$1/8$	$1/8$	$1/8$	$1/8$	$1/2$
3	$1/16$	$1/8$	0	$1/16$	$1/4$
4	$1/16$	0	$1/8$	$1/16$	$1/4$
$P_X(x)$	$1/4$	$1/4$	$1/4$	$1/4$	1

Marginal distribution of X & Y .

X	$P_X(x)$	Y	$P_Y(y)$
0	$1/4$	2	$1/2$
1	$1/4$	3	$1/4$
2	$1/4$	4	$1/4$
3	$1/4$		

$$E(X) = \sum x P_X(x) = 0 + \frac{1}{4} + \frac{2}{4} + \frac{3}{4} = \frac{6}{4} = \frac{3}{2}$$

$$E(Y) = \sum y P_Y(y) = \frac{2}{2} + \frac{3}{4} + \frac{4}{4} = 2 + \frac{3}{4} = \frac{11}{4}$$

$$E(XY) = \sum xy P_{XY}(x, y) = 10(0) \frac{1}{8} + 0(3) \frac{1}{16} + 0(4) \frac{1}{16} + 1(2) \frac{1}{8} + (1)(3) \frac{1}{8} + (1)(4) \frac{1}{16} + 2(2) \frac{1}{8} + (2)(3) 0 + (2)(4) \frac{1}{8} + 3(2) \frac{1}{8} + 3(3) \frac{1}{16} + 3(4) \frac{1}{16} = \frac{1}{4} + \frac{3}{8} + \frac{1}{2} + 1 + \frac{3}{4} + \frac{9}{16} + \frac{3}{4} = \frac{7}{4} + \frac{3}{2} + \frac{3}{8} + \frac{9}{16} = \frac{58 + 24 + 6 + 9}{16} = \frac{67}{16}$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{67}{16} - \frac{3}{2} \cdot \frac{11}{4} = \frac{67}{16} - \frac{33}{8} = \frac{67}{16} - \frac{33}{8} = \frac{67 - 33 \cdot 2}{16} = \frac{67 - 66}{16} = \frac{1}{16} = 0.0625$$

$$\sum (x^2) = \sum x^2 P_x(x)$$

$$= (0^2) \frac{1}{4} + (1^2) \frac{1}{4} + (2^2) \frac{1}{4} + (3^2) \frac{1}{4}$$

$$= \frac{1}{4} [1 + 4 + 9]$$

$$= \frac{14}{4} = \frac{7}{2}$$

$$\sum (y^2) = \sum y^2 P_y(y)$$

$$= (2^2) \frac{1}{2} + 3^2 \cdot \frac{1}{4} + 4^2 \cdot \frac{1}{4}$$

$$= 2 + \frac{9}{4} + \frac{16}{4} = 2 + \frac{1}{4} (9 + 16)$$

$$= 2 + \frac{25}{4} = \frac{8+25}{4}$$

$$= \frac{33}{4}$$

$$\text{Var}(x) = E(x^2) - (E(x))^2$$

$$= \frac{7}{2} - \left(\frac{3}{2}\right)^2 = \frac{7}{2} - \frac{9}{4}$$

$$= \frac{28-18}{8} = \frac{10}{8}$$

$$= \frac{5}{4}$$

$$\text{Var}(y) = E(y^2) - (E(y))^2$$

$$= \frac{33}{4} - \left(\frac{11}{4}\right)^2$$

$$= \frac{33}{4} - \frac{121}{16}$$

$$= \frac{11}{16}$$

$$r_{xy} = \frac{\text{cov}(x,y)}{\sqrt{\text{Var}(x)} \sqrt{\text{Var}(y)}} = \frac{7/16}{\sqrt{\frac{5}{4}} \sqrt{\frac{11}{16}}} = \frac{7/16}{(1.1180)(0.8291)}$$

$$= \frac{0.0625}{0.9269}$$

$$r_{xy} = 0.0674$$

$$r_{xy} = \frac{625}{9269}$$

calculate the equations of the lines of regression from the following data.

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

Note:

$$\bar{x} = \frac{\sum x}{n}$$

$$\bar{y} = \frac{\sum y}{n}$$

$$\text{Var}(x) = \sigma_x^2 = \frac{1}{n} \sum u^2 - \left(\frac{\sum u}{n}\right)^2$$

$$\text{Var}(y) = \sigma_y^2 = \frac{1}{n} \sum v^2 - \left(\frac{\sum v}{n}\right)^2$$

$$\text{cov}(x,y) = c_{xy} = \frac{1}{n} \sum uv - \left(\frac{\sum u}{n}\right) \left(\frac{\sum v}{n}\right)$$

Regression line of y on x.

$$y - \bar{y} = \frac{c_{xy}}{\sigma_x^2} (x - \bar{x})$$

(or)

$$y - \bar{y} = \frac{r_{xy} \sigma_y}{\sigma_x} (x - \bar{x})$$

Regression line of x on y.

$$x - \bar{x} = \frac{c_{xy}}{\sigma_y^2} (y - \bar{y})$$

(or)

$$x - \bar{x} = r_{xy} \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$\therefore u = x - \bar{x}$$

(17)

sol:

		$u = x - \bar{x}$	$v = y - \bar{y}$			
		$u = x - 4$	$v = y - 11$	u^2	v^2	uv
x	y	u	v			
1	9	-3	-2	9	4	6
2	8	-2	-3	4	9	6
3	10	-1	-1	1	1	1
4	12	0	1	0	1	0
5	11	1	0	1	0	0
6	13	2	2	4	4	4
7	14	3	3	9	9	9
$\Sigma x = 28$		$\Sigma y = 77$		$\Sigma u^2 = 28$	$\Sigma v^2 = 38$	$\Sigma uv = 31$

$$n = 7.$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{28}{7} = 4.$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{77}{7} = 11.$$

$$\sigma_x^2 = \frac{1}{n} \Sigma u^2 - \left(\frac{\Sigma u}{n} \right)^2$$

$$= \frac{28}{7}$$

$$= 4.$$

$$\sigma_y^2 = \frac{1}{n} \Sigma v^2 - \left(\frac{\Sigma v}{n} \right)^2$$

$$= \frac{38}{7} = 5.42857$$

$$C_{xy} = \frac{1}{n} \Sigma uv - \left(\frac{1}{n} \Sigma u \right) \left(\frac{1}{n} \Sigma v \right)$$

$$= \frac{31}{7} - (0)(0)$$

$$C_{xy} = 4.42857$$

Line of Regression of y on x.

$$y - \bar{y} = \frac{C_{xy}}{\sigma_x^2} (x - \bar{x})$$

$$y - 11 = \frac{3.71}{4} (x - 4)$$

$$y - 11 = 0.9275x - 3.71$$

$$0.9275x - y + 7.59 = 0.$$

(or)

$$4y - 11 = 3.71x - 14.84$$

$$3.71x - 4y = -29.16$$

Line of Regression of x on y.

$$x - \bar{x} = \frac{C_{xy}}{\sigma_y^2} (y - \bar{y})$$

$$x - 4 = \frac{3.71}{5.42857} (y - 11)$$

$$x - 4 = 0.6835y - 10.5035$$

$$0.6835y - x + 6.5035 = 0$$

(or)

$$4x - 16 = 3.71y - 40.81$$

$$3.71y - 4x = 24.81$$

- (18) calculate the correlation coefficient for the height (in inches) of father (x) and their son (y)

x	65	66	67	68	68	69	70	72
y	61	68	65	68	72	72	69	71

Ans: $r_{xy} = r_{uv} = 0.603$.

- (19) For the following data find the correlation coefficient between marks in mathematics and marks in statistics.

Mark in mathematics	25	28	35	32	31	36	29	38	34	32
Mark in statistics	43	46	49	41	36	32	31	30	33	39

Ans: $r_{uv} = -0.39$.

- (20) A study of prices of rice at Chennai & Madurai give the following data.

	Chennai (x)	Madurai (y)
Mean	19.5	17.75
S.D	1.75	2.5

Also the coefficient of correlation between the two is 0.8. Estimate the most likely price of rice

i) at Chennai corresponding to the price of 18 at Madurai.

- ii) at Madurai corresponding to price of 17 at Chennai.

Sol:

Given,

Mean, $\bar{x} = 19.5$, $\bar{y} = 17.75$.

S.D, $\sigma_x = 1.75$, $\sigma_y = 2.5$.

$\therefore r_{xy} = 0.8$.

- i). Regression line of x on y.

$$x - \bar{x} = \frac{r_{xy} \sigma_x}{\sigma_y} (y - \bar{y})$$

$y = 18$,

$$x - 19.5 = 0.8(1.75) (18 - 17.75)$$

$$x - 19.5 = \frac{1.4}{2.5} (0.25)$$

$$x - 19.5 = 0.56 (0.25)$$

$$x - 19.5 = 0.14$$

$$x = 0.14 + 19.5$$

$$\boxed{x = 19.64}$$

- ii). Regression line of y on x.

$$y - \bar{y} = \frac{r_{xy} \sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 17.75 = \frac{(0.8)(2.5)}{1.75} (17 - 19.5)$$

$$y = \frac{5}{1.75} (-2.5) + 17.75$$

$$= -\frac{5}{1.75} + 17.75$$

$$y = -2.8571 + 17.75$$

$$y = 14.8928$$

51. A statistical investigator obtain the following equations in a survey $x - y - 6 = 0$ and $0.64x - 4.08 = 0$. Here x is age of husband & y is age of wife. Find.

i). Mean of x & y .

ii). Correlation coefficient btw x & y

Sol:

Given,

$$0.64x = 4.08$$

$$x = 6.375$$

$$\bar{x} = 6.375$$

$$x - y - 6 = 0$$

$$x - 6 = y$$

$$y = 6.375 - 6$$

$$\bar{y} = y = 0.375$$

ii).

$$x - y - 6 = 0$$

$$x = (1)y + 6 \rightarrow \text{line of Regression of } y \text{ on } x$$

$$y = (1)x - 6$$

$$b_{yx} = 1$$

$$0.64x = 4.08$$

$$x = \frac{4.08}{0.64}$$

$$x = 6.375 + 0y \rightarrow \text{line of regression of } x \text{ on } y.$$

$$b_{xy} = 0$$

$$r_{xy}^2 = b_{xy} \times b_{yx}$$

$$r_{xy}^2 = 0$$

$\therefore$ Hence x and y are uncorrelated.

52. In a laboratory record of an analyze of correlation data the following results are variance of $x = 1$, the regression equations are $3x + 2y = 26$ and $6x + y = 31$. Find

i). Mean values of x and y .

ii). the standard deviation of y .

iii). Correlation coefficient btw x & y .

Sol:

$$i). \bar{x} = x = 4$$

$$\bar{y} = y = 7$$

ii). S.D of y .

$$\text{Given, Var}(x) : \sigma^2 = 1$$

(or)

$$S.D = \sigma_x = 1$$

iii)

$$\text{case (i) : } 3x + 2y = 26$$

$$3x = -2y + 26$$

$$x = -\frac{2}{3}y + \frac{26}{3}$$

$$b_{xy} = -\frac{2}{3}$$

$$6x + y = 31.$$

$$y = -6x + 31.$$

$$b_{yx} = -6$$

$$\therefore r_{xy}^2 = b_{xy} \cdot b_{yx}.$$

$$= -\frac{3}{5} \cdot -6$$

$$r_{xy}^2 = 4$$

$$r_{xy} = \sqrt{4} = -2.$$

$$\therefore \sin \theta, -1 \leq r_{xy} \leq 1.$$

case (ii)

$$3x + 2y = 26.$$

$$2y = -3x + 26.$$

$$y = -\frac{3}{2}x + \frac{26}{2}.$$

$$b_{yx} = -\frac{3}{2}.$$

$$6x + y = 31$$

$$6x = -y + 31$$

$$x = -\frac{1}{6}y + \frac{31}{6}.$$

$$b_{xy} = -\frac{1}{6}.$$

$$r_{xy}^2 = b_{xy} \cdot b_{yx}.$$

$$= -\frac{1}{6} \cdot -\frac{3}{2}$$

$$r_{xy}^2 = \frac{1}{4}$$

$$\therefore r_{xy} = \frac{-1}{2} = -0.5$$

which is in the interval $-1 \leq r_{xy} \leq 1$

ii) find σ_y :

$$\text{Formula: } \frac{\sigma_y^2}{\sigma_x^2} = \frac{b_{yx}}{b_{xy}}.$$

$$\sigma_y^2 = \frac{-3/2}{-1/6}$$

$$\sigma_y^2 = 9.$$

$$\text{S.D. } \sigma_y = 3.$$

Transformation of Random Variables:

(i) One Dimensional Random Variable

$g(x)$ is a strictly increasing.

$$f_y(y) = f_x(x) \left| \frac{dx}{dy} \right|$$

(63) Given Random variable x with density fn

$$f(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0, & \text{elsewhere.} \end{cases}$$

find pdf of $Y = 8x^3$.

Sol:

$$f_y(y) = f_x(x) \left| \frac{dx}{dy} \right|$$

$$Y = 8x^3$$

$$x = \frac{1}{2}y^{1/3}$$

$$\frac{dx}{dy} = \frac{1}{2} \cdot \frac{1}{3} y^{-2/3}$$

$$\frac{dx}{dy} = \frac{y^{-2/3}}{6}$$

$$f_v(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \frac{y^{-2/3}}{6} dx$$

$$= \frac{y^{-2/3}}{6} \times \frac{y^{1/3}}{3}$$

$$f_v(y) = \frac{y^{-1/3}}{6}$$

For, $0 < x < 1$

$$0 < \frac{y^{1/3}}{2} < 1$$

$$x < 2, \quad 0 < y^{1/3} < 2$$

raise power by 3, $0 < y < 8$

$$\therefore f_v(y) = \frac{y^{-1/3}}{6}, \quad 0 < y < 8$$

③. Two Dimensional Random Variable:

i). write down x & y in terms of u & v

(if v is not given, we take $v = y$)

$$ii) \text{ Find Jacobian } J \begin{pmatrix} x, y \\ u, v \end{pmatrix} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$iii). f(u, v) = |J| f(x, y)$$

iv) write down the limits for u & v

Let x & y be two continuous random variable with joint probability density fn.

$$f(x, y) = \begin{cases} 4xy, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}, \text{ determine}$$

Joint probability density function of $U = x^2$, $V = xy$ and hence obtain marginal probability density fn of U & V .

Sol:

$$\text{Given, } f(x, y) = \begin{cases} 4xy, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$U = x^2, \quad V = xy$$

$$u = x^2, \quad v = xy$$

$$\sqrt{u} = x, \quad v = \sqrt{u} \cdot y$$

$$y = \frac{v}{\sqrt{u}}$$

$$\text{Jacobian, } J \begin{pmatrix} x, y \\ u, v \end{pmatrix} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{1}{2} u^{-1/2} & 0 \\ -\frac{v}{2} u^{-3/2} & \frac{1}{\sqrt{u}} \end{vmatrix}$$

$$= \frac{1}{2} u^{-1/2} \cdot u^{-1/2} - 0$$

$$= \frac{1}{2u}$$

$$f(u, v) = |J| f(x, y)$$

$$= \frac{1}{2u} (1 + xy)$$

$$= \frac{1}{2u} \cdot \frac{1}{2} (u^{1/2}) \cdot \frac{v}{u^{1/2}}$$

$$\boxed{f(u, v) = \frac{2v}{u}}, \quad 0 < v^2 < u < 1$$

For, $0 < x < 1$

$$0 < u^{1/2} < 1$$

Squaring, $0 < u < 1$

For, $0 < y < 1$

$$0 < \frac{v}{\sqrt{u}} < 1$$

$$0 < v < \sqrt{u}$$

$$0 < v^2 < u$$

$$\text{Hence, } \boxed{0 < v^2 < u < 1}$$

to find Marginal Pdf:

$$f_u(u) = \int_{v=0}^{\sqrt{u}} f(u, v) dv$$

$$= \int_0^{\sqrt{u}} \frac{2v}{u} dv$$

$$= \left[\frac{2v^2}{2u} \right]_0^{\sqrt{u}}$$

$$= \frac{(\sqrt{u})^2}{u} = \frac{u}{u} = 1$$

$$\boxed{f_u(u) = 1}$$

$$f_v(v) = \int_{u=v^2}^1 f(u, v) du$$

$$= \int_{v^2}^1 \frac{2v}{u} du$$

$$= \int_{v^2}^1 2v \cdot \frac{du}{u}$$

$$= 2v [\log u]_{v^2}^1$$

$$= 2v [\log 1 - \log v^2]$$

$$= 2v (0 - \log v^2)$$

$$\boxed{f_v(v) = -4v \log v}, \quad 0 < v^2 < u < 1$$

Q. If joint PDF of (x, y) is given by $f(x, y) = x + y$, $0 \leq x, y \leq 1$. Find the pdf of $U = xy$

Q. If X & Y are independent random variables with $f(x) = e^{-x} u(x)$ and $f(y) = 3e^{-3y} u(y)$. Find $f(z)$ if $z = x/y$.

Sol:

Since x & y are independent.

$$f(x, y) = f_x(x) \cdot f_y(y) = 3e^{-3y} \cdot e^{-x}$$

$$f(x, y) = 3e^{-(x+3y)} \quad \text{for } x, y \geq 0$$

Introduce auxiliary RV, $W = Y$

$$Z = X/Y$$

$$X = ZW, \quad Y = W$$

$$J\left(\frac{x,y}{z,w}\right) = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{vmatrix}$$

$$= \begin{vmatrix} W & Z \\ 0 & 1 \end{vmatrix}$$

$$= |W - 0|$$

$$J = W$$

$$f(z,w) = |J| f_{XY}(x,y)$$

$$= |W| \times 3e^{-(z+3)w}, \quad z, w \geq 0$$

$$f(z,w) = 3w e^{-w(z+3)}$$

The pdf of z ,

$$f_z(z) = \int_0^{\infty} 3w e^{-(z+3)w} dw$$

$$\left[\because \text{Gamma Integral, } \int_0^{\infty} t^n e^{-at} dt = \frac{n!}{a^{n+1}} \right]$$

Here, $n=1$ & $a=(z+3)$

$$f_z(z) = 3 \left[\frac{1!}{(z+3)^2} \right]$$

$$f_z(z) = \frac{3}{(z+3)^2}, \quad z \geq 0$$

Sol:

Given,

$$f(x,y) = x+y, \quad 0 \leq x, y \leq 1$$

$$U = XY$$

Let us take $V = Y$

$$U = XY$$

$$X = \frac{U}{V}, \quad Y = V$$

$$J\left(\frac{x,y}{u,v}\right) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{1}{v} & -\frac{u}{v^2} \\ 0 & 1 \end{vmatrix}$$

$$J = \frac{1}{v}$$

The joint pdf of (u,v)

$$f_{UV}(u,v) = \frac{1}{|J|} f_{XY}(x,y)$$

$$= \frac{1}{|v|} \left(\frac{u}{v} + v \right)$$

For, $0 \leq X \leq 1, \quad 0 \leq Y \leq 1$

$$0 \leq \frac{u}{v} \leq 1, \quad 0 \leq v \leq 1$$

$$0 \leq u \leq v$$

$$\text{Here, } 0 \leq u \leq v \leq 1$$

The pdf of U ,

$$f_U(u) = \int_{v=u}^1 f(u,v) dv$$

$$= \int_u^1 \frac{1}{v} \left(\frac{u}{v} + v \right) dv$$

$$= \int_u^1 \left(\frac{u}{v^2} + 1 \right) dv$$

$$= \left[-\frac{u}{v} + v \right]_u^1$$

$$= [-u + 1 + 1 - u]$$

$$= 2 - 2u$$

$$f_U(u) = 2(1-u), \quad 0 < u < 1$$

18.

Sol:-

X	Y	u	v	u ²	v ²	uv
65	67	-3	-2	9	4	6
66	68	-2	-1	4	1	2
67	65	-1	-2	1	4	2
67	68	-1	-1	1	1	1
68	72	0	3	0	9	0
69	72	1	3	1	9	3
70	69	2	0	4	0	0
72	71	4	2	16	4	8
$\Sigma X = 544$	$\Sigma Y = 552$	$\Sigma u = 0$	$\Sigma v = 0$	$\Sigma u^2 = 36$	$\Sigma v^2 = 44$	$\Sigma uv = 24$

$$n = 8$$

$$\bar{x} = \frac{\Sigma X}{n} = \frac{544}{8} = 68$$

$$\bar{y} = \frac{\Sigma Y}{n} = \frac{552}{8} = 69$$

$$\text{Var}(X) = \sigma_x^2 = \frac{1}{n} \Sigma u^2 - \left(\frac{\Sigma u}{n} \right)^2$$

$$= \frac{36}{8} - \left(\frac{0}{8} \right)^2$$

$$= \frac{36}{8} = \frac{9}{2}$$

$$\text{Var}(X) = \sigma_x^2 = \frac{1}{n} \Sigma u^2 - \left(\frac{\Sigma u}{n} \right)^2$$

$$= \frac{36}{8} - 0$$

$$\text{Var}(X) = 4.5$$

$$\text{Var}(Y) = \sigma_y^2 = \frac{1}{n} \Sigma v^2 - \left(\frac{\Sigma v}{n} \right)^2$$

$$= \frac{44}{8} - 0$$

$$\text{Var}(Y) = 5.5$$

$$\text{Cov}(X, Y) = \frac{1}{n} \Sigma uv - \left(\frac{1}{n} \Sigma u \right) \left(\frac{1}{n} \Sigma v \right)$$

$$= \frac{24}{8}$$

$$\text{Cov}(X, Y) = 3$$

$$r_{xy} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}} = \frac{3}{\sqrt{4.5} \sqrt{5.5}}$$

$$= \frac{3}{4.975}$$

$$r_{xy} = 0.60302$$

19. Sol:-

X	Y	u	v	u ²	v ²	uv
25	43	-7	5	49	25	-35
28	46	-4	8	16	64	-32
35	49	3	11	9	121	33
32	41	0	3	0	9	0
31	36	-1	-2	1	4	2
36	32	4	-6	16	36	-24
29	31	-3	-7	9	49	21
38	30	6	-8	36	64	-48
34	33	2	-5	4	25	-10
32	39	0	1	0	1	0
$\Sigma X = 320$	$\Sigma Y = 380$	$\Sigma u = 0$	$\Sigma v = 0$	$\Sigma u^2 = 140$	$\Sigma v^2 = 398$	$\Sigma uv = -93$

$$n=10$$

$$\bar{x} = \frac{\sum x}{n} = \frac{320}{10} = 32$$

$$\bar{y} = \frac{\sum y}{n} = \frac{380}{10} = 38$$

$$\begin{aligned}\text{Var}(x) &= \sigma_x^2 = \frac{1}{n} \sum u^2 - \left(\frac{\sum u}{n}\right)^2 \\ &= \frac{140}{10}\end{aligned}$$

$$\text{Var}(x) = 14$$

$$\begin{aligned}\text{Var}(y) &= \sigma_y^2 = \frac{1}{n} \sum v^2 - \left(\frac{\sum v}{n}\right)^2 \\ &= \frac{398}{10}\end{aligned}$$

$$\text{Var}(y) = 39.8$$

$$\begin{aligned}C_{xy} &= \frac{1}{n} \sum uv - \left(\frac{1}{n} \sum u\right) \left(\frac{1}{n} \sum v\right) \\ &= \frac{-93}{10}\end{aligned}$$

$$C_{xy} = -9.3$$

$$\begin{aligned}r_{xy} &= \frac{C_{xy}}{\sqrt{\text{Var}(x)} \sqrt{\text{Var}(y)}} = \frac{-9.3}{\sqrt{14} \sqrt{39.8}} \\ &= \frac{-9.3}{23.6050}\end{aligned}$$

$$r_{xy} = -0.393$$